



Empowering Learners to Build Production - Ready
AI Agents

Applied Agentic AI

Foundational Content

How to study Foundational Material?

Learner Resource Guide

Foundational Content – Learner Resource Guide

The **Foundational Resource Collection** is designed to support learners who want to build stronger **conceptual clarity** around Agentic AI.

Important to know upfront

- All content in this section is **completely optional**
- **Python is NOT required for this course**
- **We do not use Python programming anywhere** in live sessions, hands-on work, or assessments

You can **skip this entire section** and still successfully complete the course.

We will also release an **AI exercise after one week**, which we strongly encourage you to attempt. It's completely okay if you're unable to work on it—we will **walk through and discuss the exercise after Week 1**.

What's Included in the Foundation resources

1. Evolution of Generative AI

(To understand what GenAI is and how it evolved)

2. Basics of Prompt Engineering

Learn how to **communicate effectively with AI tools** and shape model behavior using prompts.

You'll cover:

- How LLMs interpret instructions and context
- Types of prompts and common prompting patterns
- Prompt refinement techniques for clarity and reliability

3. Basics of Text Processing

Build intuition for how text is prepared and represented before it reaches an LLM or agent.

You'll cover:

- Fundamental concepts of Natural Language Processing (NLP)
- Typical workflow for NLP tasks
- Tokenization, part-of-speech tagging, named entity recognition, and parsing
- Text representation techniques:
 - Bag of Words
 - TF-IDF (Term Frequency–Inverse Document Frequency)
 - Word embeddings (Word2Vec)

4. Python Programming

This is an **optional, Python track** for learners who want *additional technical context* outside the core program.

You will **not** write Python, read Python, or be evaluated on Python during the course.

What it covers

- Python Fundamentals Crash course
 - Introduction to Python
 - Variables and Data Types
 - Operators
 - Lists
 - Tuples
 - Sets
 - Dictionaries
 - Practical Examples
 - Strings
 - Conditionals in Python : If-else statements
 - Loops
 - Functions
- Python for GenAI
 - NumPy
 - Pandas Fundamentals
 - Pandas : Data Analysis

- Plotting with Matplotlib

Who may find this useful

- Learners curious about implementation details
- Those planning to explore GenAI projects independently later
- Learners who want deeper technical intuition

Who can safely skip this

- Anyone focused purely on **agent design, workflows, and real-world use cases**
- Anyone uncomfortable with programming
- Anyone short on time

How to Use This Foundation

You are **not expected** to complete everything

You can:

- Skim selectively
- Use it as reference material
- Skip it entirely

Nothing here blocks your progress in the course.

Key Points

This course focuses on:

- **Agentic AI design**
- **Hands-on agent building**
- **Workflows, orchestration, tools, and real-world use cases**

The Foundational Resource Collection exists **only as optional support**—so you can learn at your own comfort level.